

THE FREE EDITION / VERSION 2

AI-Level

From LLMs to a Type II Civilization

The Five Circles and Humanity's Grand Challenges

How can artificial intelligence help us solve harder problems, build lasting capabilities, and expand humanity's possibilities?

This short book introduces five circles of progress, forty core problems, and eighteen research programs from Physics World Model. It connects today's AI with a distant horizon: a civilization that uses energy on the scale of its star.

Chengshuai Yang

September 2026

Free to read and share.

A better future begins with a problem worth solving.

From answers to capabilities

Imagine asking an AI to design a machine that repairs a water pump. A convincing answer is a beginning. Someone must make the parts, assemble the machine, test it, and keep it working. If the result succeeds, a community gains a lasting capability: restoring its water supply.

That journey is the central idea of AI-Level. Intelligence matters when it becomes something people can depend on.

What should we measure?

An AI can reason well and still fail to finish a task. It can save information without learning from it. A group of agents can exchange messages without working effectively together. AI-Level examines these differences:

- **Cognition:** solving the problem in front of the system.
- **Memory:** retaining, finding, updating, and using information over time.
- **Individual improvement:** changing methods in ways that improve later work.
- **Organization:** coordinating people, agents, tools, and checks effectively.
- **Task difficulty:** the complexity and uncertainty of the work attempted.
- **Human assistance:** the thinking and intervention people still supply.
- **Self-awareness:** recognizing capabilities, uncertainty, and limitations; this does not establish consciousness.

Measure the model together with its tools, memory, and workflow. Some systems retain state across conversations; others operate under an explicitly temporary setup. Report the system tested. A useful assessment also distinguishes a successful result, an honest inability to finish, and a false claim of completion.

The cycle behind progress

Identify a problem. Implement a solution. Validate the result. Put it to use. Then learn from what happens.

People and AI can share these tasks. Reliable evaluation requires criteria and evidence that the proposed solution cannot quietly redefine. Checks can be automated, and the checking system needs evaluation too.

A circle describes where the work takes place: software, computing hardware, physical industry, biology, or stellar infrastructure. The circles overlap and develop together. They offer a research agenda, with progress demonstrated one useful capability at a time.

Circle I: Software

Turn useful answers into reliable work

A large language model, or LLM, generates language from learned patterns. Combined with tools, memory, and a process for acting, it can form part of an AI agent. This circle asks how to make its work dependable and its improvements lasting.

Eight core grand problems

- **I-01. Memory that supports learning.** Retain useful experience and improve unfamiliar work without losing earlier abilities.
- **I-02. Dependable reasoning and action.** Finish long tasks, recover from errors, and recognize when information is missing.
- **I-03. Verified self-improvement.** Change methods and demonstrate useful gains under independent assessment.
- **I-04. Better ways to improve.** Improve how the system finds, tests, and selects changes to its methods.
- **I-05. Discovery that becomes capability.** Confirm a new finding and show that using it improves later work.
- **I-06. Trustworthy evaluation.** Detect hidden failures and shortcuts, with protected criteria and checks on the evaluator itself.
- **I-07. Organizations that learn.** Coordinate contributors and make successful methods part of ordinary practice.
- **I-08. Honest measurement.** Compare outcomes alongside uncertainty, cost, time, resources, and human assistance.

What progress would look like

A system repeatedly improves a useful tool. Independent tests confirm gains on unfamiliar tasks and check that earlier capabilities survive. The improved tool works in actual use, with the resources and human help recorded.

If useful changes are already being proposed, better evaluation may be a good place to focus. If the proposals themselves are poor, reasoning or learning may deserve attention first.

Why it matters: Dependable software can support research, education, public services, and work in every other circle.

Four physics frontiers

What could an AI scientist discover?

Physics World Model lists eighteen frontier research programs. Four connect naturally to the discovery work in Circle I. Their computational parts offer possible starting projects; the larger scientific questions also depend on the quality of theories, observations, and experiments.

L1-911. Quantum gravity

How do quantum theory and gravity fit together? One proposed task reconstructs geometric features from quantum information in a specified theoretical model. Success would demonstrate a useful reconstruction method within that model. It would be a contribution toward the larger question, rather than proof that the model describes our universe.

L1-912. Dark matter

What causes the gravitational effects attributed to dark matter? AI can help compare explanations of galaxy motions while accounting for uncertain measurements and ordinary matter. A better fit can improve an analysis; identifying dark matter's physical nature requires evidence beyond that fit.

L1-913. Dark energy

Why is cosmic expansion accelerating? A tractable task is to test methods for reconstructing expansion histories and comparing them with observations. Recovery can be measured in simulated universes where the answer is known. For our universe, researchers must report uncertainty and the assumptions behind their conclusions.

L1-914. Black-hole information

How is quantum information accounted for when a black hole evaporates? A starting task computes how information-related quantities change in a specified model. Reproducing an established result checks the method. A genuinely new conclusion needs further scrutiny and a clear statement of its scope.

A benchmark is a beginning

A benchmark is a defined task with a way to assess results. These registry pages propose tasks; they do not supply a completed solution to the full scientific problem. First establish usable data, an executable test, and a fair baseline. Then ask what a successful result would actually teach us.

The opportunity: Build a tool that helps scientists distinguish a promising explanation from one that merely fits familiar examples.

Circle II: Computing hardware

Build the machines that make intelligence possible

Every AI system depends on physical equipment. A chip design becomes useful when it can be manufactured, powered, cooled, maintained, and connected to a working system. This circle asks how intelligence can improve that whole chain.

Eight core grand problems

- **II-01. Useful chip design.** Create verified designs that meet functional, manufacturing, energy, and cost requirements.
- **II-02. Reliable fabrication.** Automate and control the production steps that turn designs into working devices.
- **II-03. Measurement and yield.** Find defects, manage process changes, and increase the share of devices that work.
- **II-04. Better devices and connections.** Develop materials, components, packaging, and links that perform useful work reliably.
- **II-05. More computation per unit of energy.** Reduce the energy required by complete systems, including memory and communication.
- **II-06. Cooling and reliability.** Remove heat and sustain useful operation over an appropriate working life.
- **II-07. Essential equipment and supplies.** Maintain access to the tools, materials, repairs, and components production requires.
- **II-08. Running on what was made.** Integrate manufactured hardware into the producing system's own working computers and verify its benefits.

What progress would look like

A team designs a component, manufactures it with acceptable quality, integrates it, and demonstrates useful performance after counting energy, equipment, and maintenance. It names the suppliers and capabilities on which it still depends.

An accessible production process may benefit most from better yield. Another project may be held back by equipment access or integration. The right first problem depends on the actual record.

A naming distinction: Circle II means computing hardware. A Type II civilization refers to stellar-scale energy use.

Six routes through hardware

Quantum systems, molecules, and materials

These six Physics World Model programs explore possible hardware capabilities. Their value depends on the application; other routes remain open.

L1-921. Fault-tolerant quantum computing

Build quantum computers that maintain useful logical operations despite errors. A starting task compares error-correction methods under realistic noise and timing constraints. Better error correction is a component milestone; a useful computing system requires much more.

L1-923. Optimal quantum control

Control quantum devices accurately despite noise and changing conditions. AI can propose control signals and test their robustness. Success in a simulator needs comparison with independently measured device performance before supporting a hardware claim.

L1-924. Quantum circuit compilation

Translate a quantum program into efficient operations on a particular machine. Check that the translation preserves the intended computation. Compare operation count, depth, time, and execution quality: improving one can make another worse.

L1-934. Molecular electronics

Explore molecules as electronic components. Predicting their behaviour is one step. Reliable contacts, reproducible manufacture, useful function, and operating lifetime determine whether a promising molecule becomes a practical device.

L1-915. Room-temperature superconductivity

Find materials that superconduct under useful operating conditions. AI can help choose candidates, but a prediction is not a discovered material. Independent physical measurements must establish the result. Pressure, stability, current, and manufacture matter alongside temperature.

L1-922. Quantum networks and repeaters

Distribute useful quantum states despite signal loss and limited memory. A starting task compares network designs and scheduling methods. Quantum communication still has physical and classical-communication requirements; it does not allow messages to travel faster than light.

Connection to the wider journey: Materials and networking advances may support industry or space systems. Their contribution must be demonstrated within those systems, alongside conventional alternatives.

Circle III: Physical industry

Make, maintain, and expand productive capacity

A productive workshop must do more than repeat one demonstration. It must handle imperfect materials, repair failures, and keep producing useful outputs.

Eight core grand problems

- **III-01. Robust robots.** Work reliably across unfamiliar objects and changing conditions.
- **III-02. Diagnosis and repair.** Identify failures, repair equipment, and verify restored function.
- **III-03. Raw materials to products.** Connect extraction, processing, and manufacture into a dependable production chain.
- **III-04. Precision manufacture.** Build accurate components, instruments, and complex machines.
- **III-05. Supplies and logistics.** Sustain material flows, transport, inventory, and recycling through disruptions.
- **III-06. Reproducing productive capacity.** Build functioning successor equipment while declaring imported parts and outside assistance.
- **III-07. Energy and infrastructure.** Maintain the power and supporting services that production requires.
- **III-08. Physical verification.** Independently measure output quality, durability, and performance under actual use.

Two frontiers in making things

L1-931. Atomically precise manufacturing: Explore ways to place atoms or reactive components accurately enough to build useful structures. A simulated placement result must eventually connect to measured physical output, throughput, and reliability.

L1-932. DNA self-assembly: Program DNA components to form useful structures and test their assembly yield. Self-assembly uses supplied parts; it does not demonstrate a system that produces its own functioning successors.

What progress would look like

A workshop produces goods, maintains equipment, and builds a functioning successor machine. Its record identifies outside supplies and human skills. Focus next on the missing capability that most limits production.

Why it matters: Stronger industrial capability can improve water systems, housing, transport, and local resilience.

Circle IV: Biology

Turn knowledge of life into healthier lives

AI can suggest explanations, design experiments, and analyse results. The biological mission connects that work to reproducible science and meaningful benefits. Health is a purpose in its own right, pursued alongside the other circles.

Eight core grand problems

- **IV-01. Causal understanding.** Discover which mechanisms drive an outcome and test what happens when they change.
- **IV-02. Reproducible experimental laboratories.** Connect planning, experiments, measurement, and learning, with important findings independently repeated.
- **IV-03. Cancer.** Improve prevention, detection, treatment, and durable control in specified cancers.
- **IV-04. Aging and healthy life.** Understand aging and test ways to preserve function and extend healthy years.
- **IV-05. Repair and regeneration.** Restore damaged tissues and organs with useful, durable function.
- **IV-06. Brain and nervous system disorders.** Improve understanding, measurement, and treatment of complex neurological conditions.
- **IV-07. Effective interventions and delivery.** Develop treatments that reach their targets and produce useful effects with acceptable adverse effects.
- **IV-08. Accessible, lasting benefit.** Make validated interventions practical and monitor their outcomes across populations.

What progress would look like

A defined intervention produces a meaningful benefit that independent work supports. Evaluation follows the stages actually reached: computation, experiments, appropriate human studies, delivery, and monitoring. Each result makes a specific claim about who benefits and under what conditions.

The first constraint may be an unreliable assay, an incomplete mechanism, weak efficacy, difficult delivery, or the time needed to observe the relevant outcome. Faster computation helps some of these steps; the evidence required still follows the claim.

Why it matters: Reducing suffering and preserving the ability to live well give this circle an immediate human purpose.

Six frontiers in living systems

Connect promising measurements to useful outcomes

The registry names five programs in life sciences and one in targeted delivery. Each offers a way to ask a narrower question within a major human-benefit mission.

L1-901. Reverse aging

Restore biological function while preserving appropriate cell identity. A younger reading on a biological-age test is a clue to investigate. It does not by itself establish rejuvenation, a longer healthy life, or a safe intervention.

L1-902. Longevity

Extend healthy life and preserve function. Models can help researchers analyse survival and design informative studies. Predicting a survival improvement and causing one are different achievements, with different evidence requirements.

L1-903. Curing cancer

Study how tumours evolve and resist treatment. Adaptive therapy is one research approach to controlling that evolution. Delaying progression, maintaining control, and achieving a cure are distinct outcomes; success in one cancer cannot establish a universal cure.

L1-904. Regeneration

Help damaged tissue regain useful function. A model may suggest how signals influence repair or scar formation. Its predictions need comparison with independently measured recovery, integration, durability, and adverse effects.

L1-905. Proteostasis

Understand how cells maintain healthy proteins and what goes wrong when proteins accumulate or malfunction. Measuring reduced protein clumping can help test a mechanism. Whether that change improves function remains a separate question.

L1-933. Targeted nanomedicine

Improve delivery of treatments to the intended tissue while reducing harmful exposure elsewhere. Measuring where a drug goes is useful. A treatment must also produce a beneficial effect and remain practical to manufacture and deliver.

The standard: Match the claim to the evidence. A better prediction, a changed biological marker, and a healthier person each represent a different kind of result.

Circle V+: Stellar infrastructure

Build a durable future beyond Earth

A Type II civilization uses energy on the scale of its star. A Dyson swarm is one proposed route: many separate orbiting structures that collect stellar energy. Early space industry could advance toward this horizon while remaining far below Type II scale.

Eight core grand problems

- **V-01. Off-world supplies.** Obtain and process materials beyond Earth while accounting for transport and imported equipment.
- **V-02. Orbital construction and growth.** Add useful productive capacity faster than it is lost through wear and failure.
- **V-03. Durable collectors.** Build energy-collecting structures with suitable lifetime and resource requirements.
- **V-04. Useful energy.** Convert, distribute, store, and use collected power effectively.
- **V-05. Waste heat.** Release heat at a rate consistent with continued operation and material limits.
- **V-06. Orbits and collisions.** Coordinate motion, manage debris, and prevent failures from spreading.
- **V-07. Long-term maintenance.** Inspect, repair, replace, and upgrade equipment when help is distant.
- **V-08. Organization across distance.** Coordinate local decisions, delayed messages, failures, and responsibilities over long periods.

What progress would look like

An orbital production system expands useful capacity over a stated period after counting losses, maintenance, energy, and imported supplies. That is an integration milestone. Reaching Type II would require a vastly greater energy scale.

The eighteen registry programs do not include a dedicated stellar-industrial challenge. Possible contributions from superconductivity and quantum networks belong beside these eight essential engineering problems. They cannot substitute for supplies, useful power, heat rejection, or repair.

Local systems must make some decisions before distant messages arrive. The need for wider coordination depends on the task and timescale, including when ordinary communication is sufficient.

Keep the purpose visible: Energy use measures one capability. It does not establish wisdom, fairness, or quality of life.

Which problem should we work on?

The answer depends on the mission. A software team may be limited by unreliable evaluation. A factory may need a precision component. A laboratory may need a trustworthy measurement. Several circles can deserve sustained effort at the same time.

Follow the constraint

Imagine an automated laboratory. First, poor planning wastes experiments. Better planning exposes inconsistent sample handling. Once handling improves, measurement quality becomes the main obstacle. The center of the work shifts as supporting capabilities become dependable.

This is what **centering on a circle** means for a particular program: focusing on the domain whose constraints now govern its next meaningful advance. "Circle time" is sustained attention to that work. It is not a fixed calendar in which all humanity must complete one domain before beginning another.

Ask what would change if a proposed obstacle disappeared. Would useful output improve, failures decline, or costs fall? If something else still determines the outcome, reconsider the priority.

Write a problem card

1. **Purpose:** Whose situation improves if this problem is solved?
2. **Constraint:** What prevents that improvement, and what evidence points to it?
3. **Contribution:** What specific capability can your team build?
4. **Demonstration:** What would count as progress or failure, and who can check it?
5. **Next use:** How can someone adopt the result and build on it?

A small team might develop a diagnostic method for one common pump failure. A useful demonstration would test unfamiliar faulty pumps, record failed repairs, and establish whether other technicians can use the method. The contribution is limited in scope and valuable in practice.

Researchers, engineers, technicians, teachers, entrepreneurs, funders, public institutions, and communities can each remove a different obstacle. A reusable dataset, a reproducible experiment, a repaired machine, or a dependable service can help many others move forward.

Choose a challenge. Find collaborators. Demonstrate what works. Give others something dependable enough to use.

Continue the journey

The five circles connect an everyday question about AI to a larger question about humanity: how far can we extend our ability to understand, build, repair, and care?

The forty core problems provide a broad map. The eighteen Physics World Model programs provide selected research directions within it. Several programs support the same core problem, and some cross circles. They are linked agendas rather than forty separate registry entries.

Explore Physics World Model

Visit the [frontier registry](#). At the September 6, 2026 review, all eighteen pages described their frontier tasks as not yet benchmarked or verified. That status concerns the registry's proposed tasks, not the absence of research in those fields.

Check the actual data and requirements before starting. Your next contribution may be documentation, a reproducible baseline, a benchmark, an independent check, an instrument, or a solution. A registry record makes work easier to find; the strength of a result comes from its evidence.

Inside the full book

AI-Level: From LLMs to a Type II Civilization develops this agenda across 32 planned chapters: intelligence levels, evidence and measurement, the engineering of each circle, detailed problem cards, competing approaches, and institutions that can sustain the work.

This free edition offers the central ideas and problems. The full book develops the reasoning and practical detail. Follow the [AI-Level book project](#) for public materials.

Sources and further reading

- [The original circle framework](#), by Chengshuai Yang.
- [NASA: dark matter and the holographic entanglement proposal](#).
- [A theoretical Page-curve calculation](#) and [IBM: quantum teleportation](#).
- [FDA: biomarkers and clinical outcomes](#) and [Wright: Dyson Spheres](#).

Find the problem. Build the capability. Demonstrate the progress. Help others take the next step.